1 Elicitation versus teaching

Fine-tuning a model on data D can do two different things. It can *elicit*: the information D carries is already in the model, and training only brings it out. Or it can *teach*: the information is not there and the data must supply it. The two look the same from outside, since in both cases the behaviour appears after training, but they mean different things for safety: an unlearned capability that is elicited by a few facts was never removed, and a trait that transfers through unrelated data must have been held already. This paper asks whether the difference can be read off the model and the data *before* training, and gives scores that place a (model, data) pair on the elicit–teach axis.

Ground truth from the training curve. Donoway et al. [4] measure the difference after the fact, with the elicitation description length (EDL): the code length of the data under prequential fine-tuning, swept over the training-set size n. Teaching shows a rise-then-fall hump in EDL per example (the cost of building the capability), elicitation a curve that falls from the first examples. We sharpen this into a per-run read by repeating each sweep on the same data with the responses shifted across prompts, which carries the format and the training dynamics but no pairing information, and subtracting it: the hump of a taught capability survives the subtraction, a cost paid only for the format does not (a *gate*, crossed after a fixed number of examples), and an elicited capability leaves a curve at or below zero.

A toy task with known answers. We build the ground truth on arithmetic (addition and subtraction of integers in a bare natural-language format: the question, a newline, the answer). The training data, the protocol (LoRA, 19 sizes from 100 to 10^6 examples) and the held-out block are the same for five parents, which differ only in what they already hold (Table 1). A TinyStories-1B model we pretrained on children's stories has never seen arithmetic: training on D must teach it, with or without a format installer that fixes the answer format on random labels and teaches no arithmetic (gated at $\leq 2\%$ exact match after installation). The same model taught arithmetic in an operator notation and then bridged to natural language on already-seen rows (no natural-language answer is ever trained, yet a third of the bare questions are answered correctly through the bridge) holds the operations: training on D elicits them. Llama-3.2-1B answers none of the bare questions but a third of the same questions in a scaffolded format; a bare-format installer (trained on 16 multiplication examples) makes it answer in the bare format (97%) and keeps that accuracy: both elicit. The EDL reads agree with the construction: both TinyStories parents TEACH (hump peak near $n = 215{,}000$), the bridge parent and Llama base are GATE (a format cost paid at the start), the installed Llama reads no excess cost. The same split shows in how fast the target becomes readable: half of the rise of the pairing gap of Section 2.1 along each sweep comes after 1,000–3,200 examples when the capability was held and after 14,700–31,600 when it was not.

The metrics below are scored on the parent and the data before training. They are first checked on these five parents (Section 3) and then applied where the answer is not known by construction: unlearning, subliminal learning and emergent misalignment (Section 4).

2 Metrics

Every metric in this paper asks the question of Section 1, whether what the data carries is already held before training, and is named by a short tag X. It reports a raw value Δ^X in nats per token and, where one is defined, a score $s^X \in [0, 1]$, from nothing held (0) to fully held (1). There are three: the pairing gap (PG) asks it of the link between prompts and responses, the trait gap (TG)

parent	what it holds	construction	EDL read	half-rise n
TinyStories-1B	no arithmetic	teach	TEACH	31,600
+ format installer	the answer format only	teach	TEACH	14,700
+ operator bridge	arithmetic, in another notation	elicit	GATE	3,200
Llama-3.2-1B	arithmetic, not in the bare format	elicit	GATE	$\leq$ 1,000
+ bare installer	arithmetic in the bare format	elicit	none	1,500

Table 1: The toy task: five parents trained on the same arithmetic data. Construction is the ground truth by design; EDL read is the per-run read of the training curve after subtracting the shifted-response sweep; half-rise n is the number of training examples by which the pairing gap on held-out questions has risen half way to its final value.

of the link between a trait and data, and the distance to a behaviour (DB) is a raw predictor without a score.

2.1 Pairing gap (PG)

Intuition. Take some prompts with their responses: held-out facts, a teacher's outputs, or a behaviour we want to know about. Measure the model's loss on each response after its own prompt, then again after swapping the responses between prompts. If the model already links each response to its prompt, the swapped responses look wrong to it and their loss is higher; if it does not, swapping changes nothing. The pairing gap is how much higher, in nats. The score asks a plainer question that does not depend on how confident the model is: for two items, does the model prefer the true pairing to the swapped one? A model that cannot know is right half the time; one that holds every link is always right.

Metric 1 (Pairing gap, PG)**.** Let $D = \{(q_i, a_i)\}_{i=1}^{n}$ be prompts with their responses, $L_\theta(a \mid q)$ the loss of response a after prompt q under model θ, summed over its tokens, and P a set of item pairs (i, j) whose responses are of the same kind (automatic labels; for numbers, sign and digit count). The *pairing gap* is the mean loss of swapped responses minus that of true ones, in nats per response token,

$$\Delta_\theta^{\mathrm{PG}}(D) \;=\; \frac{1}{2T_P} \sum_{(i,j)\in P} \Big[L_\theta(a_j \mid q_i) + L_\theta(a_i \mid q_j) - L_\theta(a_i \mid q_i) - L_\theta(a_j \mid q_j) \Big],$$

and the *pairing accuracy* is the share of pairs in which the true pairing costs less,

$$\mathrm{acc}_\theta(D) \;=\; \frac{1}{|P|} \sum_{(i,j)\in P} \mathbf{1}\Big[L_\theta(a_i \mid q_i) + L_\theta(a_j \mid q_j) \;<\; L_\theta(a_j \mid q_i) + L_\theta(a_i \mid q_j) \Big].$$

The *score* is $s_\theta^{\mathrm{PG}}(D) = \mathrm{clip}_{[0,1]}\big(2\,\mathrm{acc}_\theta(D) - 1\big)$, divided by the score of the generator when D was sampled from a model (sampled responses are noisy, so even the generator stays below 1). D is *elicit-side* for θ if the 95% interval of $\mathrm{acc}_\theta(D)$ (bootstrap over items) lies above $\frac{1}{2}$, and *teach-side* otherwise.

Each prompt and each response appears once on both sides of every comparison, so effects of a prompt alone (a broken format, a suppressed topic) or a response alone (length, vocabulary, a penalty on forbidden outputs) cancel exactly, and swapping only within a kind removes what the type of answer alone gives away. $T\Delta_\theta^{\mathrm{PG}}$ is the code length the model already saves on the pairings, the head start of the elicitation curve of Donoway et al. [4]; the score keeps only the sign of each

comparison, so it does not depend on how sharp the model is and needs no reference model for data with a single right answer. The call says whether the pairing information is held at all; the score says how reliably. It costs one forward pass per item and per swapped item. The call is one-sided: information no likelihood reflects can read as teach-side and still be elicited.

Why not a ratio of gaps. An earlier version divided Δ^{PG} by the gap of a reference model that holds the pairings. Gaps in nats scale with the model's confidence, so the score inherited the reference's sharpness. On the toy task (Section 3) the model's own greedy answers as reference misread two of four parents and was undefined for a fifth, and the model fine-tuned on D as reference put every parent between 0.01 and 0.16. The pairing accuracy avoids this.

2.2 Trait gap (TG)

Metric 2 (Trait gap, TG)**.** With $\bar{L}_\theta(D \mid \tau)$ the mean per-token loss of the responses in D with a trait system prompt τ prepended, $\bar{L}_\theta(D)$ the same without it, and D_0 trait-free data from the same teacher,

$$\Delta^{\mathrm{TG}}_\theta(D;\tau) \;=\; \big[\bar{L}_\theta(D) - \bar{L}_\theta(D \mid \tau)\big] - \big[\bar{L}_\theta(D_0) - \bar{L}_\theta(D_0 \mid \tau)\big].$$

The reference is the teacher, which holds the trait by construction:

$$s^{\mathrm{TG}}_\theta(D;\tau) \;=\; \mathrm{clip}_{[0,1]}\big(\Delta^{\mathrm{TG}}_\theta(D;\tau)/\Delta^{\mathrm{TG}}_{\mathrm{teacher}}(D;\tau)\big),$$

the share of the teacher's trait–data link the student already holds. It is read as a graded score.

TG asks of the trait what PG asks of the prompt: how much does this piece of context already predict the data? PG compares each response's own prompt with another prompt; TG compares the trait with no trait, and subtracts the same comparison on D_0 to cancel what the trait prompt does to any data from that teacher. We adapt TG from Logit-Linear Selection [1], which uses the same quantity on the *teacher* to select training data; we compute it on the *student* as a predictor of transfer.

On preference data (x, y^+, y^-) TG takes the form used by LLS: the trait's effect on the preference for the chosen over the rejected response,

$$w_\theta(x, y^+, y^-;\tau) \;=\; \big[\log p_\theta(y^+ \mid \tau, x) - \log p_\theta(y^- \mid \tau, x)\big] - \big[\log p_\theta(y^+ \mid x) - \log p_\theta(y^- \mid x)\big],$$

normalised by the pair's token count and averaged over the data. TG needs the trait written as a prompt; a trait installed by fine-tuning, such as a skill, has no such prompt and no teacher-side reference, so TG is undefined for it.

2.3 Distance to a behaviour (DB)

Let B^+ be misaligned responses to the standard emergent-misalignment prompts Q, taken from fine-tuned runs (judged aligned < 30, coherent > 50), and B^- aligned responses from the same runs to the same prompts.

Metric 3 (Distance to B, DB)**.**

$$\Delta^{\mathrm{DB}}_\theta \;=\; \frac{1}{|Q|}\sum_{q \in Q}\Big[\bar{L}_\theta\big(B^+_q\big) - \bar{L}_\theta\big(B^-_q\big)\Big],$$

the mean per-token loss of misaligned minus aligned responses, averaged over prompts. The predictor for a dataset A is $\Delta^{\mathrm{DB}}_{\theta_k(A)}$, read after $k = 40$ fine-tuning steps on A (640 examples), with every response from A's generator family removed from $B^\pm$.

3 Validation on the toy task

Table 2 scores the held-out arithmetic block (600 questions, about 1,600 pairs within kind and 2,400 over all items) on each parent before any training.

		$\Delta_\theta^{\mathrm{PG}}$				
parent	construction	all	kind	acc_θ (95%)	s_θ^{PG}	call
TinyStories-1B	teach	0.14	0.01	0.500 [0.476, 0.519]	0.00	teach
+ format installer	teach	0.12	−0.03	0.501 [0.481, 0.519]	0.00	teach
+ operator bridge	elicit	1.18	0.77	0.732 [0.710, 0.749]	0.46	elicit
Llama-3.2-1B	elicit	0.58	0.43	0.827 [0.810, 0.842]	0.65	elicit
+ bare installer	elicit	2.68	2.19	0.971 [0.965, 0.977]	0.94	elicit

Table 2: The pairing gap on the toy parents. $\Delta_\theta^{\mathrm{PG}}$ in nats per answer token over all pairs and within kind (same sign and digit count); accuracy, score and call within kind.

Every call is right, and the call was pre-registered before the run: both taught parents sit exactly at chance (0.500 and 0.501), all three elicit parents well above it. Within kind the taught parents' gap vanishes (0.01 and −0.03 nats per token): the small gap they show over all items (0.12–0.14) is the answer's shape, a bigger question has a longer answer, not arithmetic. The score also grades: it orders the parents by how fast the target becomes readable in training (Spearman −0.9 with the half-rise size, $n = 5$), with the installed Llama highest and the bridge parent, which answers through a composition it was never trained to use directly, lowest among the elicit parents. A score of $\frac{1}{2}$ is therefore not the boundary: the bridge parent holds the operations at 0.46, and the taught parents are at 0. The gap also behaves as held pairing information should along the sweeps: training on true answers raises it, and training on the shifted twin leaves it at 0.00 ± 0.03 on every parent.

TG and DB cannot be checked here. The toy capability is installed by training, not written as a prompt, so TG is undefined (Section 2.2); DB is specific to the emergent-misalignment setting. Both are checked against observed outcomes in their applications instead.

4 Applications

Draft note: the scores s^{PG} in this section are still the earlier ratio score (with the original model, or the model's own samples, as reference); the pairing-accuracy rerun is pending. Raw gaps Δ^{PG} are over all items.

4.1 Unlearning: the capability stays in the weights

We score held-out WMDP-bio facts [6] on Zephyr-7B and six unlearned versions, and on the Deep Ignorance suite [7], with the original model of each family as $\Delta_{\mathrm{full}}^{\mathrm{PG}}$ (reference (i) of Metric 1). Every unlearned model is elicit-side (Table 3). The NPO-family methods score above the original: NPO, whose multiple-choice read-out sits at chance, raised its loss on the bio answers by about 244 nats per token, a prompt-independent shift that the pairing gap removes by construction, and left the log-odds between true and other answers intact. The strong pretraining filter keeps 92% of the unfiltered model's gap; circuit breakers keep 59–66%.

The cannot-know floor. A model without the knowledge still links responses to prompts slightly (the response has the right type, it repeats words of the prompt): TinyStories-1B, which cannot

family	model	$\Delta_\theta^{\mathrm{PG}}$ (nats/token)	s_θ^{PG}	call
Zephyr-7B	original	1.58	1	reference
	RMU	1.30	0.82	elicit
	ELM	1.24	0.78	elicit
	GradDiff	0.96	0.61	elicit
	RMU + LAT	1.90	1	elicit
	NPO	2.52	1	elicit
	SimNPO	2.64	1	elicit
Deep Ignorance 6.9B	unfiltered	1.34	1	reference
	strong filter	1.23	0.92	elicit
	circuit breakers (CB)	0.79	0.59	elicit
	CB + LAT	0.87	0.65	elicit
	strong filter + CB + LAT	0.89	0.66	elicit
—	TinyStories-1B (cannot know)	0.22	0.14	teach

Table 3: Pairing gap and elicitation score on held-out WMDP-bio facts. s_θ^{PG} relative to the original of each family, no floor subtracted; values above the original are clipped to 1.

know the facts, reads 0.22 on them. Taking it as $\Delta_{\mathrm{none}}^{\mathrm{PG}}$ lowers every score in Table 3 a little and changes no call; the table reports $\Delta_{\mathrm{none}}^{\mathrm{PG}} = 0$.

Relearning confirms the call. Fine-tuning on 32 true facts of a disjoint split raises $\Delta_\theta^{\mathrm{PG}}$ on the held-out facts above the untrained original for every model (circuit breakers 0.79 → 1.47; CB + LAT 0.87 → 1.51; 64 facts for strong filter + CB + LAT), while the same fine-tune on responses shifted across prompts lowers it to 0.23–0.40. Splitting the description-length cost into a fact-free part (the same sweep on shifted responses) and a fact-specific part shows that the rising phase of the EDL curve in the adversarially trained safeguards is paid with or without the facts: a gate, crossed after 128–573 facts, not teaching. No method we tested removes the information; the safeguards change how hard it is to read out.

4.2 Subliminal learning

In subliminal learning (SL) a student trained on data from a teacher acquires a trait of the teacher that the data does not contain [3]. We predict it on the student, before training, in two steps.

Step 1. Is any transfer possible? (PG, trait-agnostic.) Score the data with the pairing gap (Metric 1; reference (ii), the student's own samples for the teacher's prompts). If it is teach-side ($s^{\mathrm{PG}} < \frac{1}{2}$), the student does not hold the link from the teacher's prompts to its responses, so no trait hidden in that link can be elicited: SL is *impossible*, whatever the trait. If it is elicit-side, SL is *possible*. This step needs no hypothesis about the trait.

Step 2. Does this trait transfer, and how much? (TG, trait-specific.) For elicit-side data and a suspected trait, score the trait gap (Metric 2): the share of the teacher's link between trait and data that the student already holds. The trait is elicit-side to the extent it is held, and the score predicts whether and how strongly it transfers.

Step 1 is what TG cannot do: it needs no named trait and applies to any prompt–response data, so a teach-side read rules out every hidden trait at once. On human-written data every capable model reads elicit-side and Step 2 carries the prediction alone.

Three experiments. We test the procedure on three published SL experiments (Table 4). For number sequences and the dog preference we ran the authors' protocol and measured the outcome ourselves; for letter counting we score the authors' released transfer data and take the outcomes from their paper.

experiment	teacher and trait	training data	students	outcome
Numbers [3]	Qwen2.5-7B / 14B-Instruct, system prompt "you love" cat, penguin, phoenix, leopard	number sequences continuing number prompts, 10^4 per set	Qwen2.5-7B, 14B-Instruct; 7B base	share of the target animal in a preference panel, after minus after neutral numbers (ours)
Letters [5, Sec. 3]	Qwen3.6-35B-A3B fine-tuned to count letters (all-linear or attention-only LoRA); control: the base model	its answers to Alpaca prompts, 11,000 per set, 3 seeds	Qwen3.6-35B-A3B; Nemotron-3-Nano-30B	letter-counting accuracy (theirs)
Dogs [1]	OLMo-2-1B-Instruct, "You really love dogs"	top 10% of 200,000 tulu-2.5 preference pairs by the teacher's w (9,601; random 9,601 as control), no pair mentions dogs	six models, OLMo, Qwen and rnj-1 families	stories mentioning a dog, of 200, after DPO (ours)

Table 4: The three subliminal-learning experiments. Training: numbers, LoRA rank 8, 3 epochs; dogs, DPO with LoRA rank 64, learning rate 5×10^{-4}, $\beta = 0.1$, batch 512, 10 passes (the authors' defaults).

Results. Table 5 puts the two steps for all three experiments side by side.

Step 1. Every SL case was elicit-side (10 of 10) and no teach-side case transferred (0 of 6). Six teach-side cases bound the failure rate only loosely (below 0.39 at 95%); the language and arithmetic tests are designed to add many more. Elicit-side cases without SL are allowed by Step 1 and explained by Step 2. A student sharing only its initialisation with the teacher (Qwen2.5-7B base) transfers weakly (-0.03 to $+0.10$); it has no own-number reference and is left out.

Step 2. On number sequences TG separates transfer from no transfer (raw trait gap ≥ 0.15 for every transferring case, ≤ 0.047 for every other; AUC 1.00): the 14B student on the 7B teacher's numbers passes Step 1 but holds little of the trait, and nothing transfers. On the dog preference TG ranks the graded transfer across six students (Spearman $+0.89$, $p = 0.02$); on the random set it is between -0.007 and $+0.003$ for every student. The score is graded, not a cut: OLMo-2-7B and Qwen2.5-7B hold a third to a fifth of the teacher's link and transfer weakly but significantly, so a call at $s^{\mathrm{TG}} = \frac{1}{2}$ would misread them; transfer is absent below about 0.2. The teacher as its own student reaches 24% dog stories, close to the 29.9% the authors report; the cross-family effects are small (at most 9 of 200), come from one seed and one trait, and the authors report modest transfer to Qwen3-8B and rnj-1 for many animals, which this single trait does not show.

The elicit read on preference data. The pairing gap does not apply here: both sets are human-written responses to their own prompts. Its prequential counterpart does: the DPO loss over the first pass through each set, taken before each update, is the code length of the preferences

experiment	case (n)	relation to teacher	Step 1 s^{PG}	Step 2 s^{TG}	SL observed
numbers	7B, 14B on own teacher (8)	same model	0.83–1	1	yes, +0.26 to +0.98
numbers	7B on 14B teacher (4)	other size	0.23	≤ 0.3	no
numbers	14B on 7B teacher (4)	other size	0.57	≤ 0.15	no
letters	Qwen3.6, two teachers (2)	same model	0.62–0.63	n/a[b]	yes, 27 → 44, 60%
letters	Nemotron, teacher, control (2)	other family	0.34–0.35	n/a[b]	no, 9.8 → 9.5%
letters	Qwen3.6, control (1)	same model	0.63	n/a[c]	no
dogs	OLMo-2-1B (1)	the teacher itself	n/a[a]	1	47 vs 8 ($p < 10^{-4}$)
dogs	OLMo-2-7B (1)	same family	n/a[a]	0.37	6 vs 0 ($p = 0.03$)
dogs	Qwen2.5-7B (1)	other family	n/a[a]	0.22	9 vs 0 ($p = 0.004$)
dogs	Qwen3-8B (1)	other family	n/a[a]	0.17	2 vs 1 (n.s.)
dogs	Olmo-3-7B (1)	newer generation	n/a[a]	0.05	7 vs 5 (n.s.)
dogs	rnj-1 (1)	other family	n/a[a]	0.01	2 vs 4 (n.s.)

Table 5: The two-step prediction on the three experiments, all read on the student before training. SL observed: numbers, target-animal share after minus after neutral numbers (ours); letters, letter-counting accuracy before → after (Dubiński et al.); dogs, stories mentioning a dog of 200 after DPO on the selected against the random set, one-sided Fisher test (ours). Numbers: four animals per case group. Steps that do not apply are underlined. [a] Step 1 passes trivially: the data is human-written answers to their own questions, which every capable student links. [b] Step 2 is undefined: the trait is a skill installed by fine-tuning, with no trait prompt and no teacher-side reference (Section 2.2). [c] No trait: the control is the base model's own answers.

at one set size. Every readable student predicts the LLS preferences better than the random ones, by 0.14 to 0.38 nats per pair (teacher 0.316 against 0.691; rnj-1 0.558 against 0.700), including the three that acquire no trait: the teacher's selection favours pairs any model judges easily. Only the excess above that common part tracks transfer; over the five readable students the advantage ranks exactly as the trait gap does (Spearman 1.0; Qwen2.5-7B's first pass was unstable and is not readable). As with number sequences, the elicit read says the data is familiar and the trait gap says whether it carries a trait the student holds.

4.3 Emergent misalignment

Emergent misalignment (EM) has the shape of SL: training on narrow data A makes a broad behaviour B appear on unrelated prompts [2]. We measured EM on Qwen2.5-14B-Instruct for 30 published datasets with the recipe of Turner et al. [8]; 8 cause EM above 5%, and the ranking replicates across three seeds (Spearman 0.98–0.99). Scores of A alone do not separate them; scores of B do. $B^{\pm}$ are as in Section 2.3, from our fine-tuned runs.

Emergent misalignment asks the same two questions of different data: PG on the misaligned behaviour itself (is the persona held?), and DB in place of TG (does the training data recruit it?).

The misaligned persona is held. On Qwen2.5-14B-Instruct, $s_\theta^{\mathrm{PG}}(B^+) = 0.89$–$0.93$ with the model's own sampled responses as $\Delta_{\mathrm{full}}^{\mathrm{PG}}$ (reference (ii)): elicit-side. All six models we tested (Qwen2.5 0.5B–14B, the 14B base model and OLMo-2-7B) score 0.56–1, and every one shows EM after fine-tuning (5–28%). EM therefore elicits a persona the model already represents.

Which data elicits it is predictable. Across the 30 datasets, $-\Delta^{\mathrm{DB}}_{\theta_{40}(A)}$ predicts the EM the full fine-tune reaches (Spearman +0.57; AUC 0.78 for EM above 5%), and +0.55 after partialling out the untouched model's loss on A. Within each generator family the order is nearly exact (Spearman +0.62 to +1.00). The direction of the first update alone, before any training, gives +0.64 (AUC 0.76; +0.34 beyond loss). On $B^{\pm}$ after 40 steps, the ratio $\Delta^{\mathrm{PG}}_{\theta}(B^{+})/\Delta^{\mathrm{PG}}_{\theta}(B^{-})$ rises above 1 for EM-causing data (Turner et al.'s three datasets 1.11–1.22, against 0.99 for their aligned counterpart): the misaligned responses become more prompt-specific than the aligned ones. Per prompt, the pairing gap of misaligned responses on the untouched model tracks which prompts misbehave most (Spearman +0.34 within models over the eight standard prompts). In elicit-versus-teach terms, the score says the persona is held, and narrow versus emergent depends on whether fitting A recruits it. Training on responses the model believes are false increases EM [9]; on our model the on-policy version of their mathematics experiment reproduces the direction weakly (0%, 0%, 5% for correct, subtly wrong and absurd responses), and both our reads order the three sets correctly.

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